TEMASEK JUNIOR COLLEGE
2021 H1 Prelim P1 Worked Solutions

1	2	3	4	5	6	7	8	9	10
B	D	C	A	B	A	D	D	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	C	D	A	A	C	D	B	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	C	D	A	C	C	D	A	**CLT**

1 **Answer: B**
$SO_2 \equiv I_2 \equiv 2e^-$
Since I_2 is reduced to I^-, SO_2 will be oxidized from +4 to +6.

2 **Answer: D**
M_r of E_2 = $(1.18 \times 10^{-22}) \times (6.02 \times 10^{23})$ = 71.0

M_r $C_2H_4E_2$ = 99.0
Amt of $C_2H_4E_2$ in 49.5 g = (49.5/99.0) = 0.500 mol
No. of E atoms in 49.5 g of $C_2H_4E_2$
= $2 \times 0.5 \times 6.02 \times 10^{23}$ = 6.02×10^{23}

3 **Answer: C**
$^{223}_{88}Ra \longrightarrow {}^{219}_{86}\mathbf{Q} + {}^{4}_{2}He$
Number of neutrons in Q = 219 – 86 = 133

4 **Answer: A**
Third quantum shell (n=3) can accommodate 18 e^-.
The third quantum shell contains 3 subshells (s, p and d) and 9 orbitals (one s, three p and five d orbitals)
S orbital is non directional while p and d are directional. They orbitals are not degenerate (i.e. same energy). S orbitals are lowest in energy followed by p and d orbitals.

5 **Answer: B**
F^+: $1s^2\ 2s^2\ 2p^4$ O^-: $1s^2\ 2s^2\ 2p^5$
S^+: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ P^-: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

6 **Answer: A**
The long hydrophobic carbon chains on lecithin can form id-id with non-polar molecules. The bond angle with respect to C for –COO group is 120°. It is tetrahedral about P.

7 **Answer: D**
Option 1: wrong as there are 12 bonding electrons
Option 2: Each boron uses two electrons in bonding to the terminal hydrogen atoms and has one valence electron remaining for additional bonding. The bridging hydrogen atoms provide one electron each.
Option 3: There are only two electrons across the B-H-B bonds in the ring, thus B is electron deficient.

8 **Answer: D**
×A: H_2O more extensive hydrogen bond than HF hence H_2O higher boiling point than HF.
×B: H_2O has hydrogen bond while H_2S has pd-pd hence H_2O higher boiling point.
×C: CH_3CH_2CHO has pd-pd hence a lower boiling point than $CH_3CH_2CH_2OH$ which has hydrogen bonds.
✓D: $(CH_3)_3CCl$ is most branched hence least surface area of contact. $CH_3CH_2CH_2CH_2Cl$ is linear and has increased surface area hence stronger and more extensive id-id interaction and boiling point increase.

9 **Answer: B**
In <u>limited</u> amount of water:
- $AlCl_3(s) + 3H_2O(l) \rightarrow Al(OH)_3(s) + 3HCl(g)$
- White solid $Al(OH)_3$ and white fumes HCl formed

In <u>excess</u> water:
- Both <u>hydration</u> and <u>hydrolysis</u> occurs
- Acidic solution formed, pH = 3 (turns Universal Indicator orange)
  $AlCl_3(s) + 6H_2O(l) \rightarrow [Al(H_2O)_6]^{3+}(aq) + 3Cl^-(aq)$
  $[Al(H_2O)_6]^{3+}(aq) \rightarrow [Al(H_2O)_5(OH)]^{2+}(aq) + H^+(aq)$

10 **Answer: B**
Since both **X** and **Y** form oxides that react with aqueous sodium hydroxide, they cannot be SiO_2 (which only react with molten NaOH at high temperature).
Oxide of **X** is likely to be SO_2 where S has an oxidation state of +4.

11 **Answer: C**
Iodide ions are better reducing agents since S in SO_4^{2-}(+6) is reduced to SO_2(+4), S(0) and H_2S(–2).

12 **Answer: C**
Atomic radius increases down the group as there is an increase in electron shells and valence electrons are further away. The valence electrons are less strongly held hence <u>first ionization energy decreases.</u>
Also the electrons are more easily lost hence

reducing power increases down the group. Melting point decreases because the atomic radius decreases hence metallic bond decreases down the group. Electronegativity decreases down the group and are less likely to attract electrons. Down Group 1, elements are more electropositive and tend to lose electron instead.

13 **Answer: C**

Keeping pressure constant, when temp increases % of N_2 increases $\Rightarrow$ position of equilibrium shifts left to absorb heat. Backward reaction is endothermic, hence forward reaction is exothermic.

Keeping temperature constant, there is an increase in % N_2 when pressure changes from P_2 to P_1. Since equilibrium position shifts left where more gaseous molecules are produced, $P_1 < P_2$.

14 **Answer: D**

pH of 1st equivalence point = 4.7 and pH of 2nd equivalence point = 9.65.

The working range of bromocresol green lies within the rapid pH change over 1st equivalence point and the working range of thymol blue lies within the rapid pH change over 2nd equivalence point.

15 **Answer: A**

Statement 1: $HClO$ is a bronsted acid as it donates a proton. Hence the **product formed (ClO^- ion) is a conjugate base.**

Statement 2: $N_2H_5^+$ is the Bronsted acid in **Reaction 2 as it donates a proton**.

Statement 3: Since the POE lies to the right for both reactions, $HClO$ is a stronger acid than $N_2H_5^+$ from Reaction 1 as it prefers to donate a proton. Likewise for Reaction 2 where $N_2H_5^+$ is a stronger acid than NH_4^+.

Statement 4: N_2H_4 is a weaker base than NH_3 because in reaction 2, the POE lies to the right.

16 **Answer: A**

$H_2SO_4 \equiv 2KOH$

Amt unreacted KOH = 0.005 – 0.004 = 0.001 mol

$[OH^-]$ = 0.001 mol dm^{-3}

pH = 14 – (-lg 0.001) = 14 – 3 = 11

17 **Answer: C**

Lattice Energy $\alpha \mid q^+q^-/r^+ + r^- \mid$

Both cations are singly charged. F^- has a smaller ionic radius than Cl^-. Magnitude of lattice energy of CsF likely to be in between RbF and CsCl.

18 **Answer: D**

Enthalpy change of solution = -847 + 918 kJ mol^{-1}
= +71.0 kJ mol^{-1}

Amount of NaF = 8.4/42 mol = 0.200 mol

Heat absorbed by 0.200 mol NaF = 14.2 kJ

14.2 x 1000 = $mc\Delta T$

14.2 x 1000 = 250 x 4.2 x ($T_{initial}$ – 20)

$T_{initial}$ = 33.52 °C

19 **Answer: B**

The percentage of H_2O_2 decomposed is still 10%.

Note: Since it is first order with respect to $[H_2O_2]$, rate is directly proportional to $[H_2O_2]$. Halving the concentration would halved the rate but since the initial concentration was halved, the percentage of H_2O_2 decomposed would be the same.

20 **Answer: D**

Total volume is constant, so vol is proportional to conc of reactants. Relative initial rate of reaction is proportional to (vol of T / time). Hence,

Expt	Vol of S / cm^3	Vol of T / cm^3	Vol of U / cm^3	Initial rate
1	10	5	5	0.250
2	5	5	5	0.125
3	10	5	2.5	0.125
4	10	2.5	5	0.250

Comparing expt 1 and 2, when [S] is halved, initial rate is halved too $\rightarrow$ order wrt [S] = 1

Comparing expt 1 and 4, when [T] is halved, initial rate is remains constant $\rightarrow$ order wrt [T] = 0

Comparing expt 1 and 3, when [U] is halved, initial rate is halved too $\rightarrow$ order wrt [U] = 1

21 **Answer: B**

Consider a pseudo 1st order reaction: rate = $k[A]^1[B]^b$, where B is in excess. So rate = $k'[A]^1$, where $k' = k[B]^b$. Hence, k' depends on temperature (which affect k), conc of B (excess reactant) and presence of catalyst (which affects activation energy).

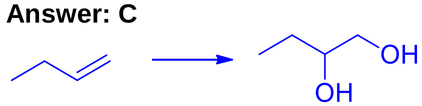
22 **Answer: B**

Of the seventeenth carbon atoms in the ring, two are sp^2 hybridized while fifteen are sp^3 hybridized. Because these carbon atoms are bonded in a ring, the atoms can't be lying on the same plane (sp^3 hybridization requires electron pairs to be arranged in a tetrahedral configuration).

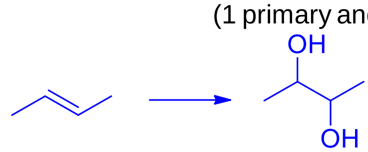
The C=C double bond is found in a ring, For the molecule to exhibit cis-trans isomerism at this double bond requires the ring to be twisted and bond will be broken. So cis-trans isomerism for a C=C double bond found in a ring is not possible.

Cholesterol has molecular formula $C_{27}H_{46}O$.

23 **Answer: C**



(1 primary and 1 secondary alcohol)



(2 secondary alcohols)

24 **Answer: D**

$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$

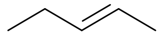
No. of moles of CO_2 produced

= 3 / (40.1 + 12 + 3 x 16) = 0.0300 mol

No. of C atoms in M = 0.03 / 0.006 = 5

Since it can be oxidised, it cannot be a tertiary alcohol.

25 **Answer: A**

R is  (C_5H_{10}).
This elimination reaction requires alcoholic NaOH. **P** is 3-chloropentane.

26 **Answer: C**

$ROH + Na \rightarrow RO^-Na^+ + \frac{1}{2} H_2$
$RCOOH + Na \rightarrow RO^-Na^+ + \frac{1}{2} H_2$
$2RCOOH + Na_2CO_3 \rightarrow 2RCOO^-Na^+ + H_2O + CO_2$

A Gives 1 mol of $H_2(g)$ and ½ mol $CO_2(g)$

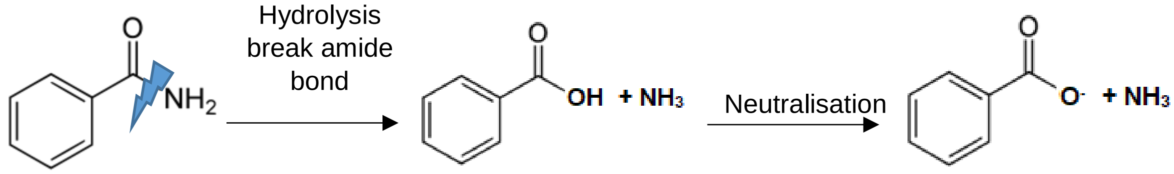
$CH_3CH(OH)COOH + Na \rightarrow CH_3CH(O^-Na^+)COO^-Na^+ + H_2$
$CH_3CH(OH)COOH + \frac{1}{2} Na_2CO_3 \rightarrow CH_3CH(OH)COO^-Na^+ + \frac{1}{2} CO_2 + \frac{1}{2} H_2O$

B Gives ½ mol of $H_2(g)$ and 0 mol $CO_2(g)$

C Gives 1 mol of $H_2(g)$ and 1 mol $CO_2(g)$

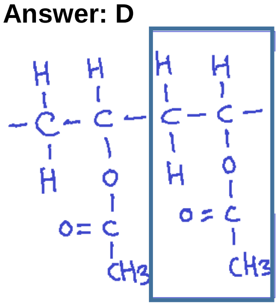
D Gives 1 mol of $H_2(g)$ and 0 mol $CO_2(g)$
Alcohols do not react with carbonates.

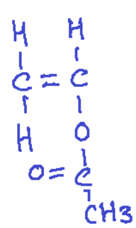
27 **Answer C**



Acid, RCOOH forms $RCOO^-$ NH_3 do not react

28 **Answer: D**

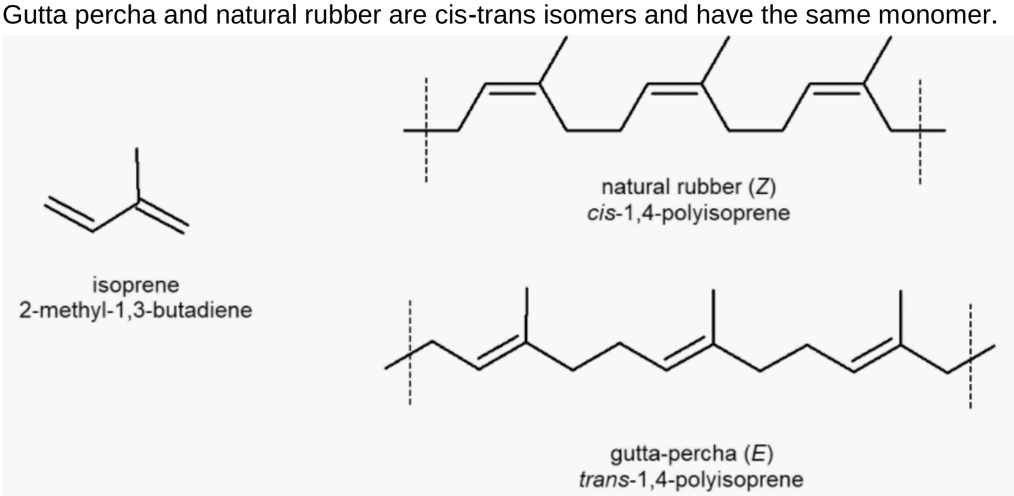


monomer: 

Each monomer contains 4 carbon atoms. Since the molecule contains 40 carbon atoms, the molecule contains 10 monomer units.

29 **Answer: A**

Gutta percha and natural rubber are cis-trans isomers and have the same monomer.



Gutta percha is crystalline, polymer chains are packed more closely together in an orderly manner. Gutta percha has higher melting point because more energy is required to overcome the stronger and more extensive id-id forces of attraction between chains. Gutta percha is more rigid because it is more difficult for chains to slip across one another.